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The singleton set is always —

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Answer (Please select your correct option)

☐ disconnected

☐ polygonally disconnected

☒ connected

☐ None of these



Any polygonally connected set either it is open or not, is always _____

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Answer (Please select your correct option)

☒ connected

☐ disconnected

☐ a region

☐ None of these



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For the function $f(x,y) = \frac{xy}{x^2+y^2}$, the limit of $f(x,y)$ as $(x,y) \rightarrow (0,0)$ along the line $y=x$ is ———

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Answer (Please select your correct option)

☐ $\frac{1}{2}$ ☐ $-\frac{1}{2}$ ☐ 0☐ 1

If f and g are continuous functions at the point X_0 in $\mathbb{R}^n$, then $\frac{f}{g}$ is continuous at X_0 provided that _____

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Answer (Please select your correct option)

☐ $g(X_0) = 0$

☒ $g(X_0) \neq 0$

☐ $f(X_0) = g(X_0) = 0$

☐ None of these



Suppose $U_0 \in \mathcal{T}$ is a limit point of $\mathcal{T}$, g is continuous at U_0 and f is continuous at $X_0 = G(U_0)$. Then _____ is continuous at U_0 .

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Answer (Please select your correct option)

☒ $f \circ g$

☐ $\frac{f}{g}$

☐ $\frac{g}{f}$

☐ None of these

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A set $A \subset \mathbb{R}$ of real numbers is _____ if there exists a real number $M \in \mathbb{R}$, such that $x \leq M$ for every $x \in A$.

Answer (Please select your correct option)

☐ bounded below☐ uniformly continuous☐ bounded above☐ None of these

A function f is _____ on a subset S of $\mathbb{R}^n$ if S is contained in an open set on which $f_1, f_2, \dots, f_n$ are continuous

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Answer (Please select your correct option)

☒ continuously differentiable

☐ uniformly convergent

☐ piecewise convergent

☐ None of these



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A critical point at which a function attains its minimum value among all points where it is defined is called _____

Answer (Please select your correct option)

☐ global maximum☒ global minimum☐ supremum☐ None of these

The rate of change of $f(x,y)$ in the direction of a unit vector is called the _____ of $f(x,y)$

Answer (Please select your correct option)

☐ differential

☒ directional derivative

☐ mixed derivative

☐ None of these



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A function of the form $L(X) = m_1 x_1 + m_2 x_2 + \dots + m_n x_n$ for some constants m_i 's is called _____

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Answer (Please select your correct option)

☒ linear function☐ quadratic function☐ constant function☐ None of these